SESD-Assignment-5

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2503A52L20.

Case Study 2 – Library Management System

Develop a system where students can search, borrow, and return books. The librarian can manage inventory, due dates, and fines.

Deliverables: Backlog with Search, Borrow, Return, Notifications, Fine Management.

1. Aim

To design a Library Management System that allows students to search, borrow, and return books while librarians manage inventory, due dates, notifications, and fines.

2. Problem Statement

Develop a Library Management System where students can search for available books, borrow books, and return them before the due date. Librarians can manage book inventory, monitor transactions, and calculate overdue fines.

3. Product Backlog

Backlog ID	User Story / Feature	Priority
PB01	Search books by title, author, or category	High
PB02	Check book availability	High
PB03	Borrow available books	High
PB04	Return borrowed books	High
PB05	Add, update, and remove books	High
PB06	Track book issue dates and due dates	High
PB07	Send notifications before due dates	Medium
PB08	Notify students about overdue books	Medium

PB09	Calculate fines for late returns	High
PB10	View and manage fine payment status	Medium

4. UML Class Diagram

Insert the UML Class Diagram provided by the instructor here.

The diagram represents the relationships between Book, Librarian, Transaction, MemberRecord, Student, Faculty, Bill, and different book categories.

5. Sprint Planning

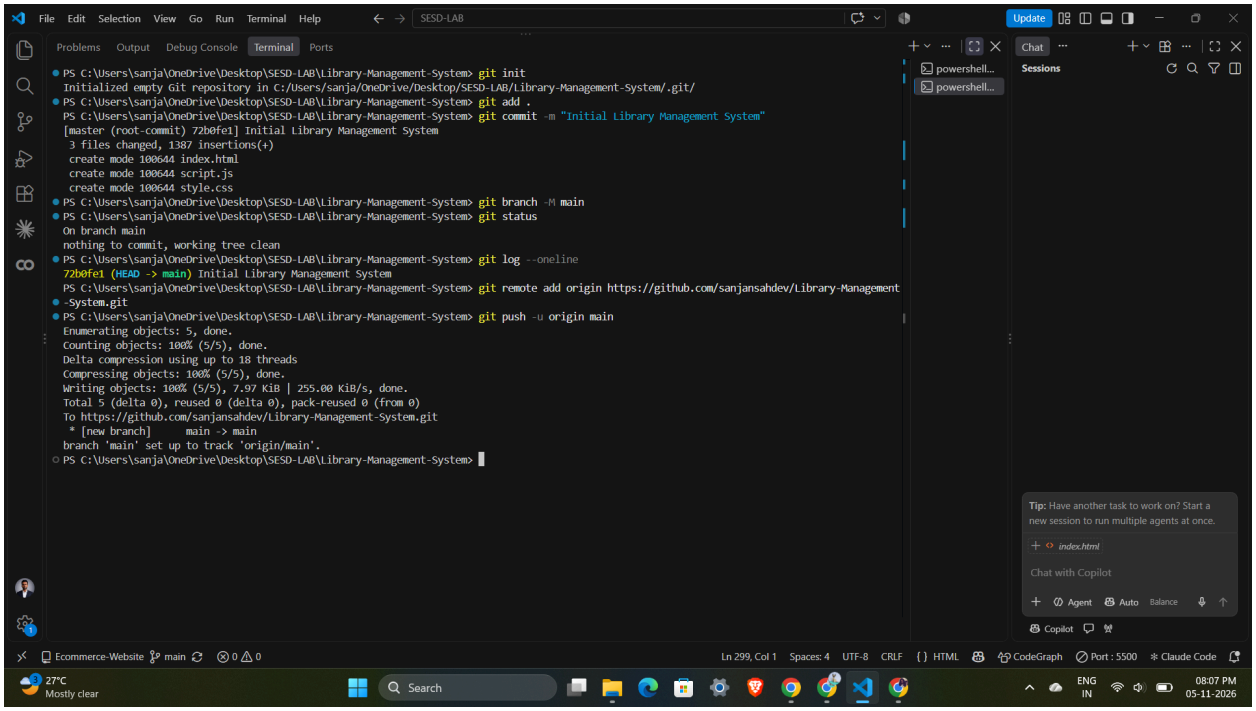
Sprint	Features
Sprint 1	Book Search and Inventory Management
Sprint 2	Book Borrowing and Returning
Sprint 3	Due Date Tracking and Notifications
Sprint 4	Fine Management and Testing

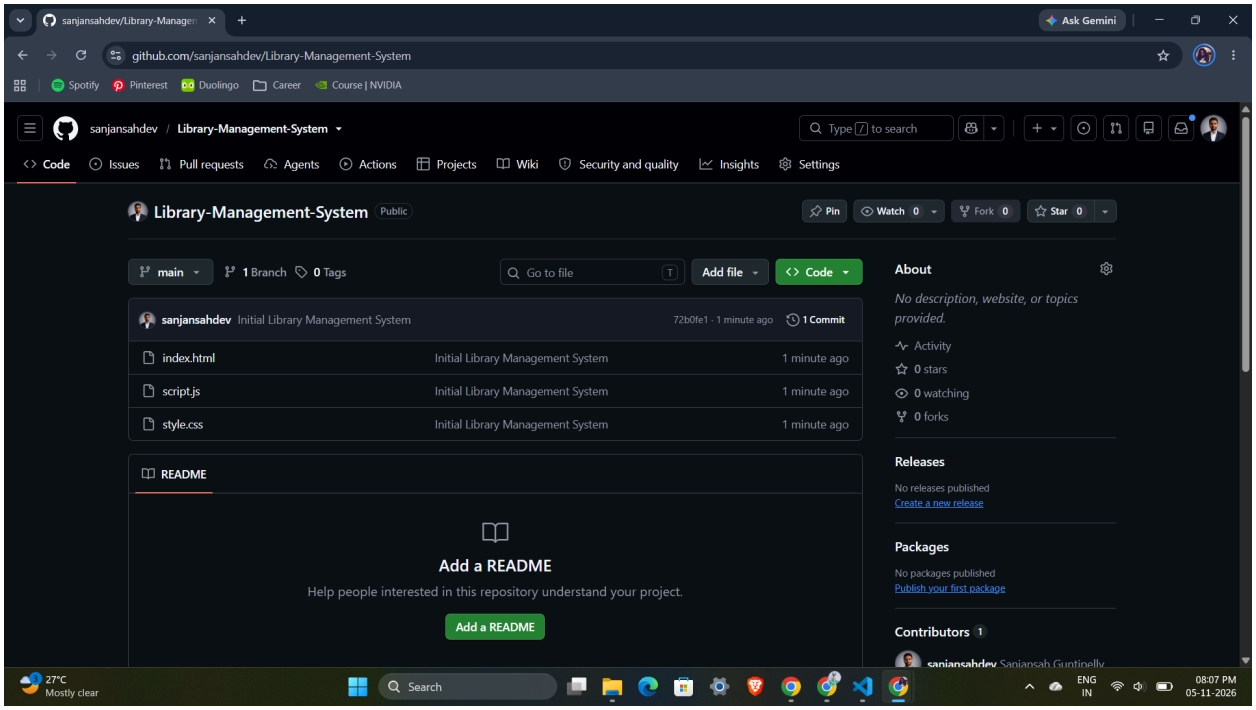
6. Result

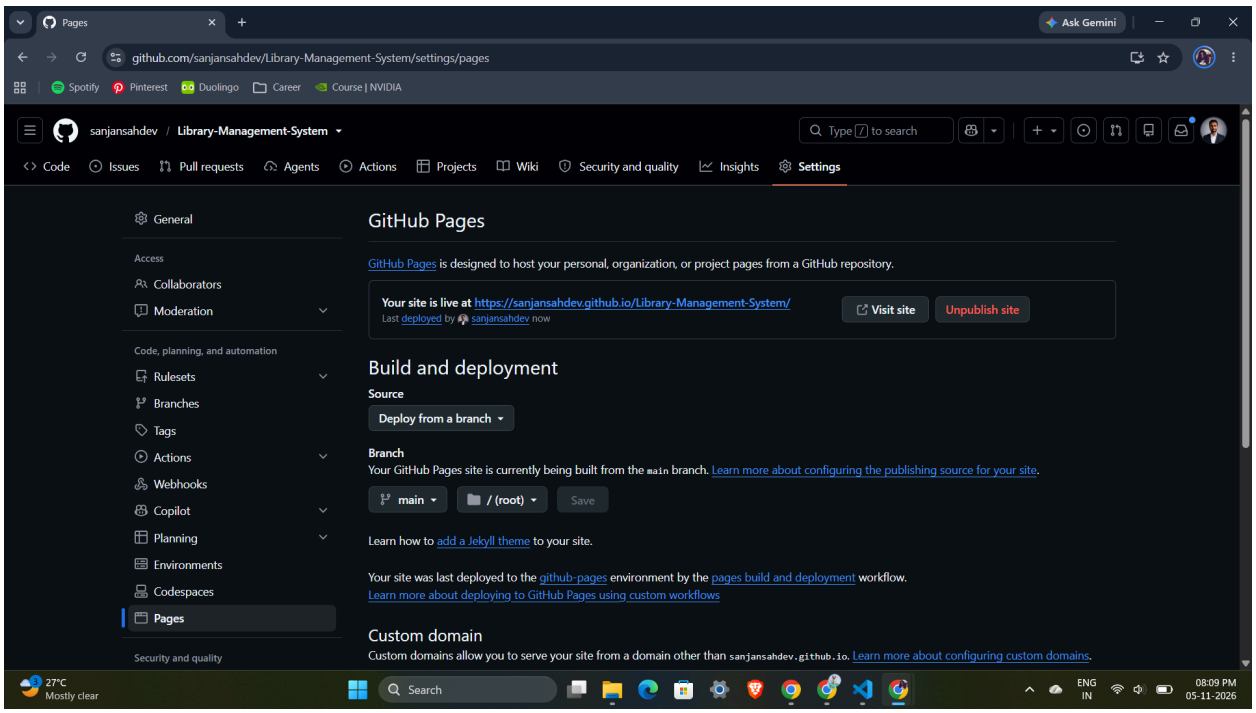
Successfully designed a Library Management System using a Product Backlog and UML Class Diagram, covering book searching, borrowing, returning, notifications, and fine management.

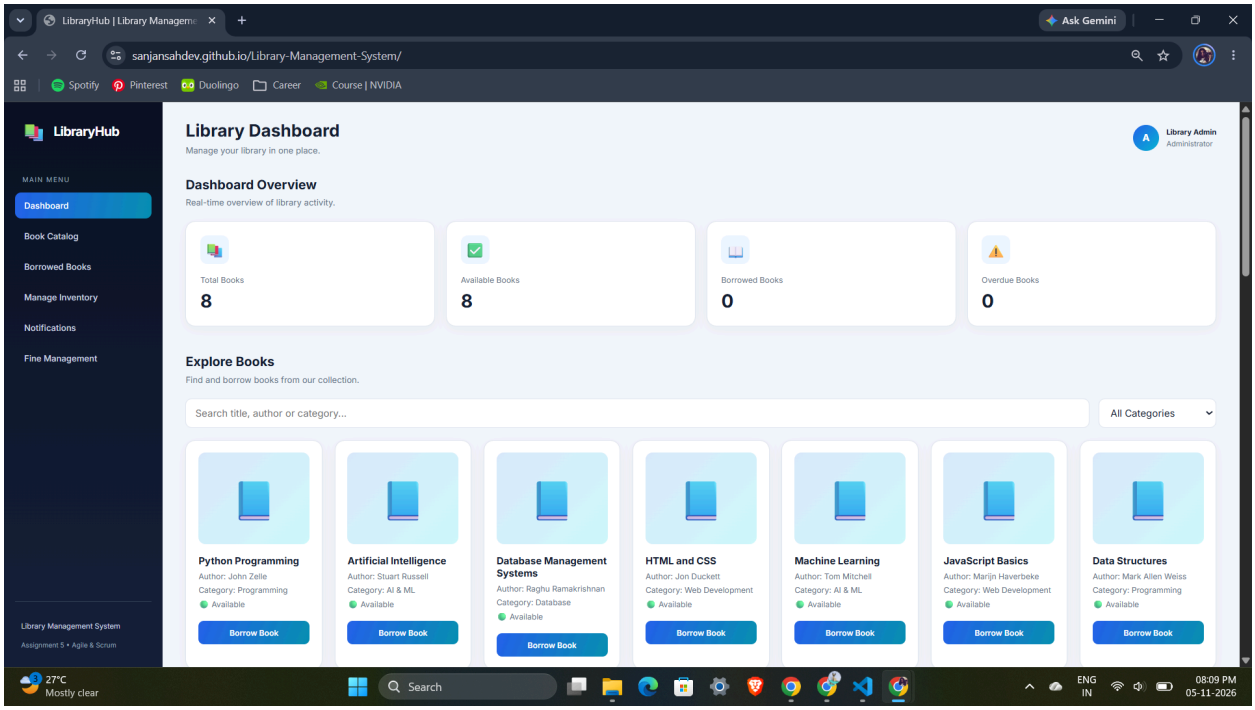
Live Website at : https://sanjansahdev.github.io/Library-Management-System/

Output Screenshots :




sanjansahdev/Library-Manager
Ask Gemini
github.com/sanjansahdev/Library-Management-System
Spotify
Pinterest
Duolingo
Career
Course | NVIDIA
sanjansahdev / Library-Management-System
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Library-Management-System
Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
Add file
Code
sanjansahdev Initial Library Management System
72b0fe1 · 1 minute ago
1 Commit
index.html
Initial Library Management System
1 minute ago
script.js
Initial Library Management System
1 minute ago
style.css
Initial Library Management System
1 minute ago
README
Add a README
Help people interested in this repository understand your project.
Add a README
About
No description, website, or topics provided.
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
27°C
Mostly clear
Search
ENG
IN
08:07 PM
05-11-2026


Pages
Ask Gemini
github.com/sanjansahdev/Library-Management-System/settings/pages
Spotify
Pinterest
Duolingo
Career
Course | NVIDIA
sanjansahdev / Library-Management-System
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
General
Access
Collaborators
Moderation
Code, planning, and automation
Rulesets
Branches
Tags
Actions
Webhooks
Copilot
Planning
Environments
Codespaces
Pages
Security and quality
GitHub Pages
GitHub Pages is designed to host your personal, organization, or project pages from a GitHub repository.
Your site is live at https://sanjansahdev.github.io/Library-Management-System/
Last deployed by sanjansahdev now
Visit site
Unpublish site
Build and deployment
Source
Deploy from a branch
Branch
Your GitHub Pages site is currently being built from the main branch. Learn more about configuring the publishing source for your site.
main
/ (root)
Save
Learn how to add a Jekyll theme to your site.
Your site was last deployed to the github-pages environment by the pages build and deployment workflow.
Learn more about deploying to GitHub Pages using custom workflows
Custom domain
Custom domains allow you to serve your site from a domain other than sanjansahdev.github.io. Learn more about configuring custom domains.
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Search
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LibraryHub | Library Management
Ask Gemini
sanjansahdev.github.io/Library-Management-System/
Spotify
Pinterest
Duolingo
Career
Course | NVIDIA
LibraryHub
MAIN MENU
Dashboard
Book Catalog
Borrowed Books
Manage Inventory
Notifications
Fine Management
Library Management System
Assignment 5 • Agile & Scrum
Library Dashboard
Manage your library in one place.
Library Admin
Administrator
Dashboard Overview
Real-time overview of library activity.
Total Books
8
Available Books
8
Borrowed Books
0
Overdue Books
0
Explore Books
Find and borrow books from our collection.
Search title, author or category...
All Categories
Python Programming
Author: John Zelle
Category: Programming
Available
Borrow Book
Artificial Intelligence
Author: Stuart Russell
Category: AI & ML
Available
Borrow Book
Database Management Systems
Author: Raghu Ramakrishnan
Category: Database
Available
Borrow Book
HTML and CSS
Author: Jon Duckett
Category: Web Development
Available
Borrow Book
Machine Learning
Author: Tom Mitchell
Category: AI & ML
Available
Borrow Book
JavaScript Basics
Author: Marijn Haverbeke
Category: Web Development
Available
Borrow Book
Data Structures
Author: Mark Allen Weiss
Category: Programming
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